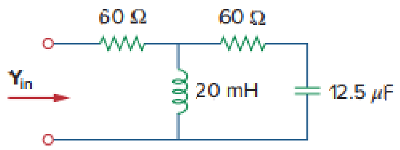
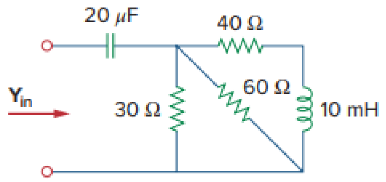
Problem

At ω = 103 rad/s, find the input admittance of each of the circuits in Fig. 9.74.

Figure 9.74:



(a)



(b)

Step-by-step solution

Step 1 of 6

Refer to figure 9.74 given in the textbook.

(a)

Calculate the impedance of inductor Z_L.

$$Z_L = j\omega L$$

Substitute 10^3 for ω, and 20×10^{-3} for *L*.

$$Z_L = j\times10^3\times20\times10^{-3}$$

$$Z_L = j20\ \Omega$$

Calculate the impedance of capacitor Z_C.

$$Z_C = \frac{1}{j\omega C}$$

Substitute 10^3 for ω, and 12.5×10^{-6} for *C*.

$$Z_C = \frac{1}{j\times10^3\times12.5\times10^{-6}}$$

$$Z_C = -j80\ \Omega$$

Step 2 of 6

Calculate the impedance seen from the same end from which the admittance is seen.

$$Z_{in} = 60 + \left[(Z_L) \| (60 + Z_C)\right]$$

Substitute $j20$ for Z_L, and $-j80$ for Z_C.

$$\begin{aligned} Z_{in} &= 60 + \left[(j20) \| (60 - j80)\right] \\ &= 60 + \frac{j20 \times (60 - j80)}{j20 + 60 - j80} \\ &= 60 + \frac{j1200 + 1600}{60 - j60} \\ &= (63.333 + j23.333)\ \Omega \end{aligned}$$

Step 3 of 6

Calculate the admittance Y_{in}.

$$Y_{in} = \frac{1}{Z_{in}}$$

Substitute $(63.333 + j23.333)$ for Z_{in}.

$$\begin{aligned} Y_{in} &= \frac{1}{(63.333 + j23.333)} \\ &= (0.014) - j(0.005122) \\ &= 0.0148\angle -20.22^\circ \\ &= 14.8\angle -20.22^\circ\ \text{mS} \end{aligned}$$

Therefore, the admittance Y_{in} is $\boxed{14.8\angle -20.22^\circ\ \text{mS}}$.

Step 4 of 6

(b)

Calculate the impedance of inductor Z_L.

$$Z_L = j\omega L$$

Substitute 10^3 for ω, and 10×10^{-3} for *L*.

$$\begin{aligned} Z_L &= j \times 10^3 \times 10 \times 10^{-3} \\ &= j10\ \Omega \end{aligned}$$

Calculate the impedance of capacitor Z_C.

$$Z_C = \frac{1}{j\omega C}$$

Substitute 10^3 for ω, and 20×10^{-6} for *C*.

$$\begin{aligned} Z_C &= \frac{1}{j \times 10^3 \times 20 \times 10^{-6}} \\ &= -j50\ \Omega \end{aligned}$$

Step 5 of 6

Calculate the impedance seen from the same end from which the admittance is seen.

$$Z_{in} = Z_C + \left[30 \| 60 \| (40 + Z_L)\right]$$

Substitute $j10$ for Z_L, and $-j50$ for Z_C.

$$Z_{in} = -j50 + \left[20 \| (40 + j10)\right]$$

$$Z_{in} = -j50 + \left[\frac{20 \times (40 + j10)}{20 + 40 + j10}\right]$$

$$Z_{in} = -j50 + \left[\frac{800 + j200}{60 + j10}\right]$$

$$Z_{in} = -j50 + 13.5135 + j1.081$$

$$Z_{in} = (13.51 - j48.92)\ \Omega$$

Step 6 of 6

Calculate the admittance Y_{in}.

$$Y_{in} = \frac{1}{Z_{in}}$$

Substitute $(13.51 - j48.92)$ for Z_{in}.

$$\begin{aligned} Y_{in} &= \frac{1}{(13.51 - j48.92)} \\ &= (5.2466 \times 10^{-3}) + j0.1899 \\ &= 19.704\angle 74.55^\circ \text{ mS} \end{aligned}$$

Therefore, the admittance Y_{in} is $\boxed{19.704\angle 74.55^\circ \text{ mS}}$.